PesaPap

Customer Growth, Retention and Value Analytics

Track B: Fintech and Mobile Money

Big idea

Grow with better customers, not just more customers.

Robin Kinyua

PLP Data Analytics Capstone | October 2026

The growth question

BUSINESS PROBLEM

Where should PesaPap focus acquisition, retention and engagement?

Stakeholder

Head of Growth

PesaPap has grown, but the team needs to know which users create the most value, which channels bring stronger customers, and where engagement weakens over time.

Decision

Prioritise acquisition spend, retention campaigns and customer engagement.

Questions this analysis answers

- Which acquisition channels bring the highest-value users?
- Which customer segments contribute the most transaction value?
- Where does cohort activity weaken over time?
- Which products and counties drive transaction activity?
- Which actions should the growth team prioritise next?

Data and analytical approach

METHOD

A reproducible end-to-end workflow across four related tables.

01

Prepare

Synthetic fintech data
8,000 users

02

Process

Quality audit +
reusable cleaning

03

Analyse

EDA, SQL, KPIs,
stats, cohorts

04

Share

Looker dashboard +
reports + slides

Core dataset

8,000 users **70,824** transactions **12** months **4** related tables

Python • pandas • NumPy • SQLite • SQL • SciPy • Matplotlib • Excel • Looker Studio

KPI overview

SCORECARD

The cleaned dataset shows strong customer activity and transaction scale.

8,000

Total customers

7,522

Active customers

94.02% active rate

70,824

Total transactions

KES 111.18M

Total value

Average transaction value

KES 1,569.78

Average customer value

KES 13,897.26

What this means

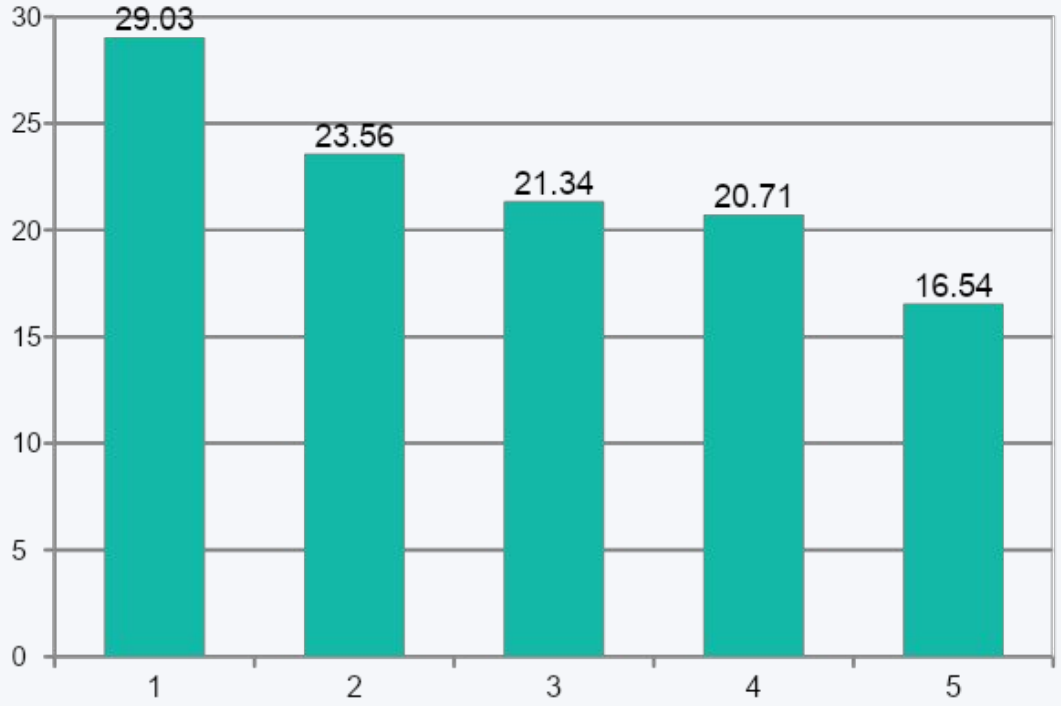
PesaPap already has broad customer activity. The bigger opportunity is improving the quality and longevity of that activity.

Referral wins on customer value; Send Money drives product value

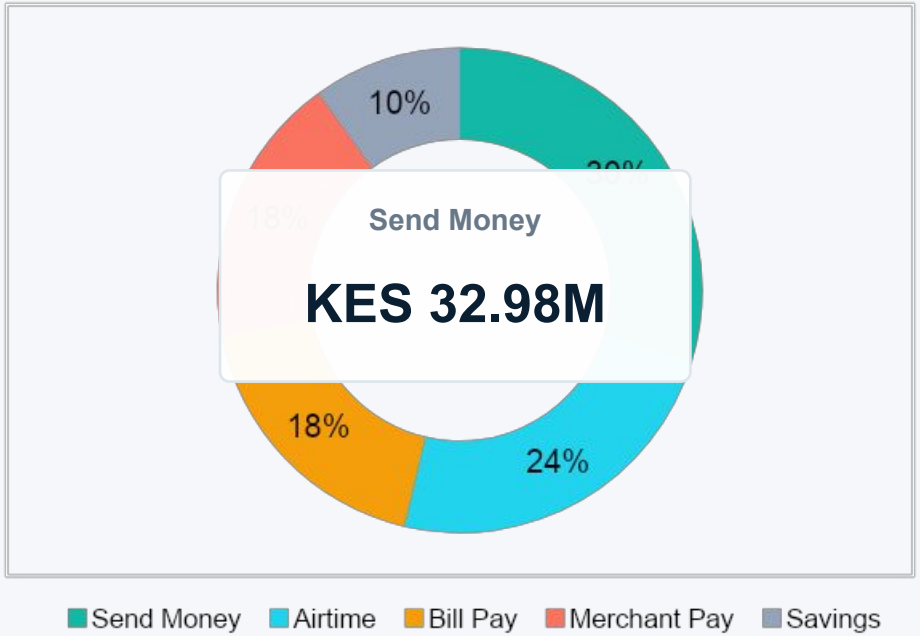
GROWTH DRIVERS

Channel quality and product usage tell the same growth story from different angles.

Transaction value by acquisition channel (KES M)



Product share of transaction value

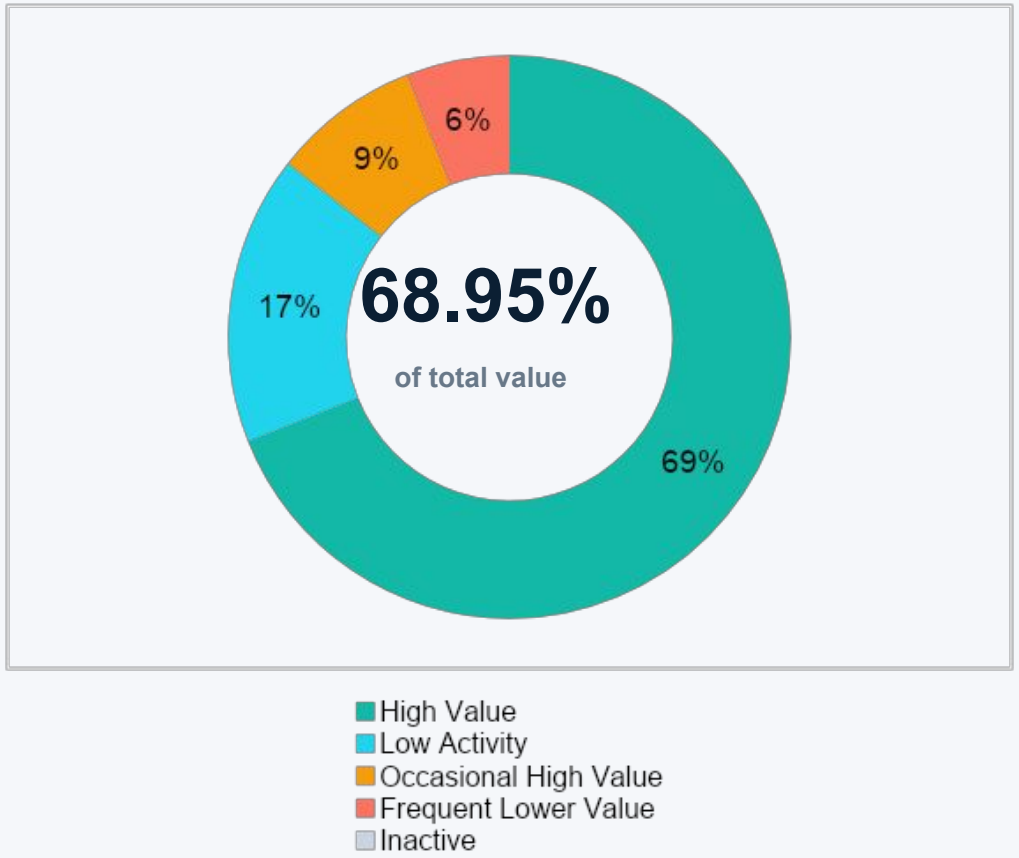


Referral users also have a higher median customer value: KES 13,212.50 vs KES 11,084.50 for Social users.

Value is concentrated in the High Value segment

SEGMENTATION

A large low-activity base creates a clear engagement opportunity.



High Value

3,221 customers

KES 76.66M

68.95% of total value

Low Activity

3,129 customers

KES 18.49M

Largest engagement opportunity

Inactive

478 customers

KES 0

Priority reactivation pool

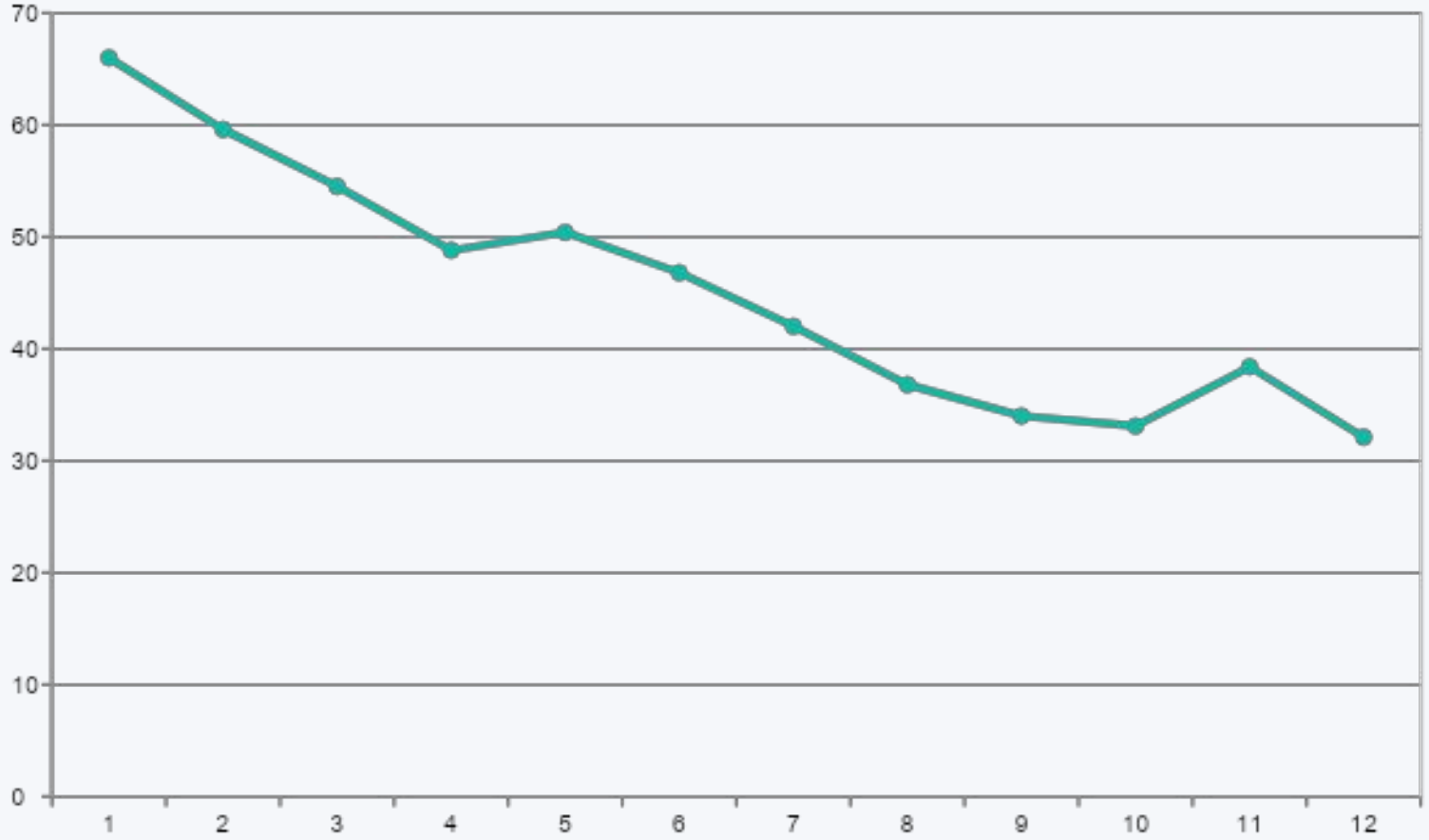
Growth action

Protect high-value customers, while moving low-activity users toward more frequent use.

Retention weakens as customers move further from signup

RETENTION

The July 2025 cohort falls from 66.0% active in month 0 to 32.1% by month 11.



Month 0

66.0%

Month 11

32.1%

-33.9 pts

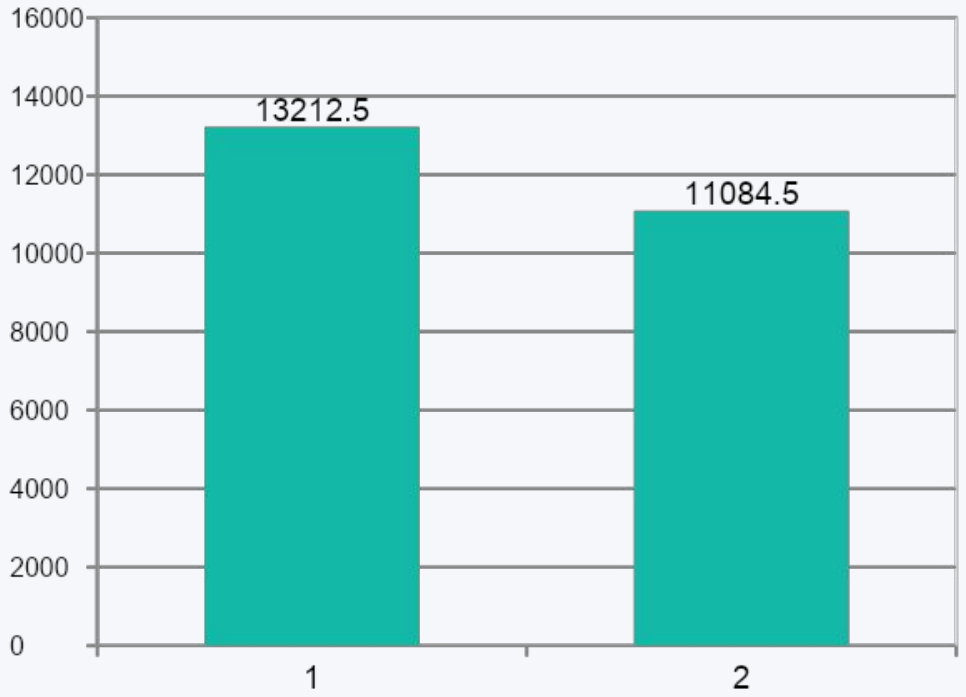
Action

Trigger lifecycle campaigns several months after signup.

Referral customers show significantly higher value than Social

STATISTICAL EVIDENCE

The difference is statistically significant and commercially meaningful.



Mann-Whitney U

1,945,284

p-value

6.24×10^{-8}

Decision

Reject H_0

Referral and Social users do not have the same customer-value distribution. Referral also has the higher median value.

Business takeaway: give Referral more acquisition attention, while validating CAC, retention and scalability before shifting spend.

Act on quality, engagement and retention

RECOMMENDATIONS

A focused growth plan can increase customer lifetime value without relying only on new-user growth.

1. **Acquire** — Prioritise Referral while monitoring acquisition cost and scalability.
2. **Protect** — Reward High Value customers who contribute 68.95% of total value.
3. **Engage** — Move Low Activity customers toward more frequent transactions.
4. **Reactivate** — Target the 478 inactive customers with reminders or incentives.
5. **Retain** — Launch lifecycle campaigns several months after signup.

Expected impact

Higher customer lifetime value

More frequent transactions

Stronger retention

Final message

Grow with better customers - and keep them active for longer.